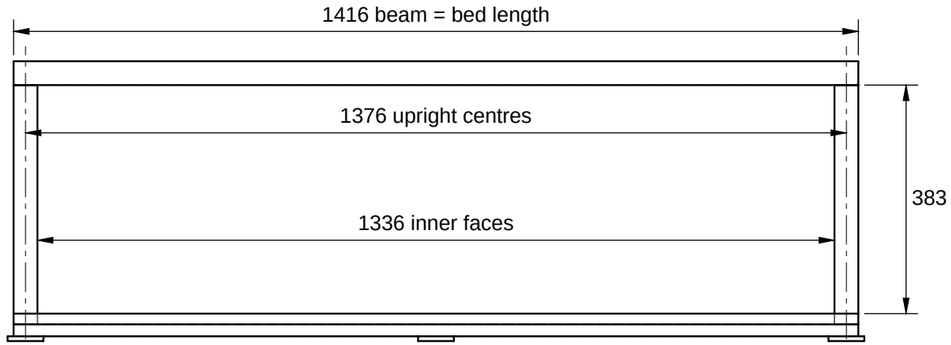
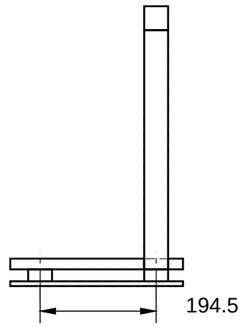



**FRAME, FRONT (1:10)**



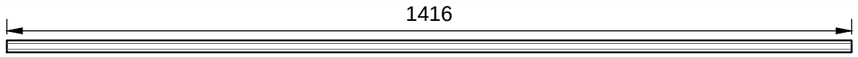
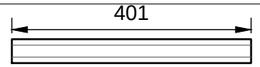
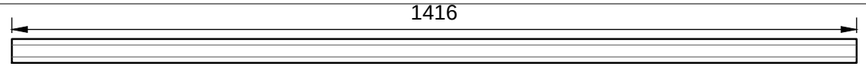
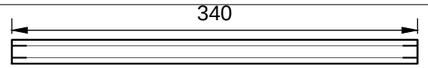
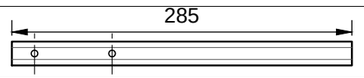
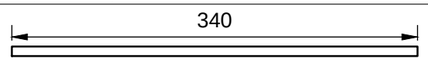
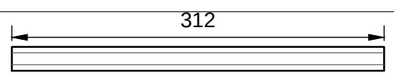
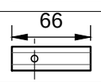
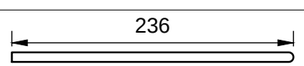

**RIGHT (1:10)**

**Frame set-up (BUILD_GUIDE 4.5; tolerances quoted)**

Base rails: centre distance (RIGHT view) +/-0.5 at both ends; diagonals equal within 1. Upright inner faces (FRONT) +/-0.5, equal at the bottom and the top [Certain, CAD]. Bed top to beam bottom +/-0.5, equal at both ends [Certain, CAD]. Uprights stand on the back base rail through notches in the bed (D-03); the beam sits on the uprights, flush with their outer faces. Joints (L brackets, back joining plates) are NOT in the CAD: BUILD_GUIDE 4.3 [Guessing on the exact layout].

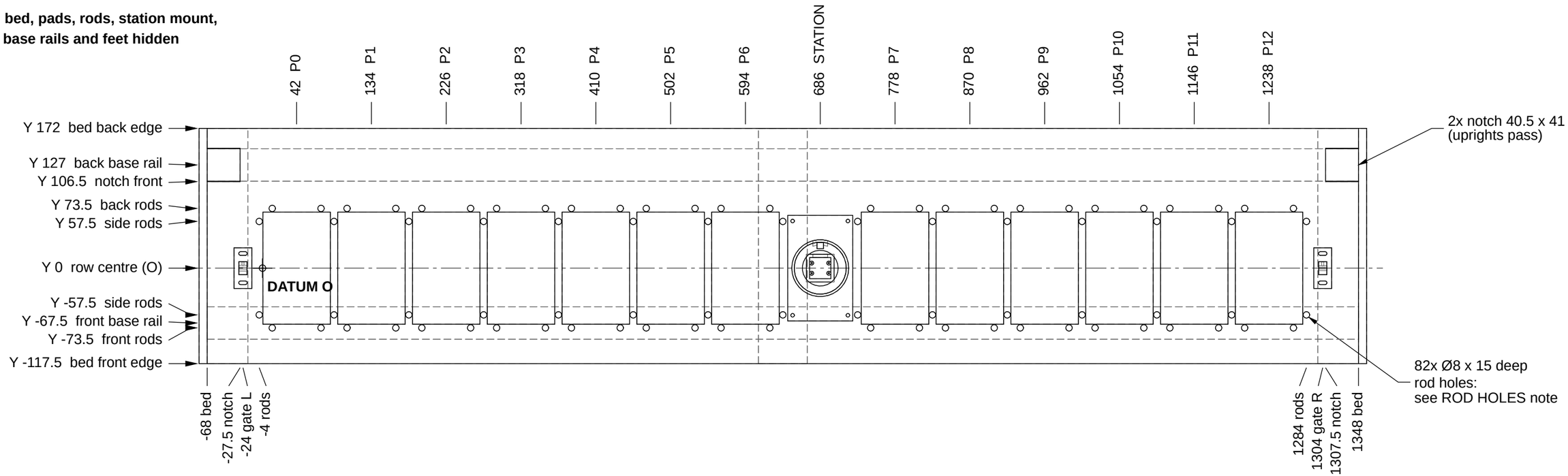
**CUT LIST (cad/out/cut_list.csv; extrusion cuts +/-0.5 with square ends, BUILD_GUIDE 4.1)**

Extrusions: 20-series, slot 6 (Misumi HFS5, BOM.csv)

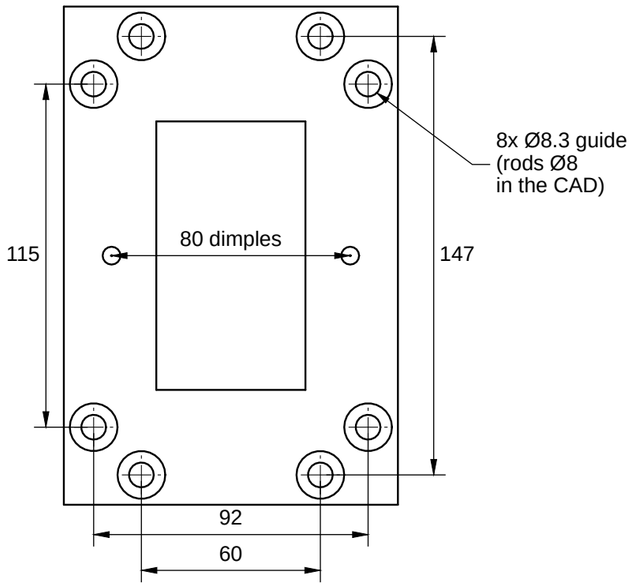
| Item | Bar (length from the CAD; scale per row) | Profile / stock | Use, ends, holes | Qty | Tolerance |
|---|---|---|---|---|---|
| C-01 |   | 2040 | base rail, lying flat under the bed; square ends | 2 | +/-0.5 |
| C-02 |   | 4040 | upright; square ends | 2 | +/-0.5 |
| C-03 |   | 4040 | X beam, on top of the uprights; square ends | 1 | +/-0.5 |
| C-04 |   | 2040 | Z spine; BOTH ends: 2x M5 tapped, the two centre bores (20 apart), for 2x M5x12 button heads (bottom) and 2x M5x16 (top), BUILD_GUIDE 7.1, 7.3 | 1 | +/-0.5 |
| C-05 |   | 2020 | head drop arm; 2x Ø5.5 THRU across the arm (machine X), 19 and 84 from the end in head_foot's socket: match-drill (BUILD_GUIDE 7.5) | 1 | +/-0.5 |
| C-06 |   | T8x8 | T8x8 Z screw, length below the motor face: cut the motor's 400 mm screw; 311-356 works (cut_list.csv) | 1 | 311-356 |
| C-07 | | 1 mm Al sheet | probe flag blank: D-06 | 1 | D-06 |
| C-08 |   | 2020 | Phase 0 jig post; cut square | 2 | +/-0.5 |
| C-09 |   | 2020 | Phase 0 jig arm stub; cross hole 19 from one end, match-drilled through head_foot | 1 | +/-0.5 |
| C-10 | | 1 mm Al sheet | carry-sensor flag: D-06 | 1 | D-06 |
| C-11 |   | 5/16 in dowel | guide rod; dome the top R2-3 (ROD_DOME_R; BUILD_GUIDE 5.4), chamfer the bottom lightly | 82 | PROPOSED +/-0.5 |

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-02 of D-17** |
| **Frame and extrusion cut list** | SCALE 1:5, 1:10 (as noted) |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

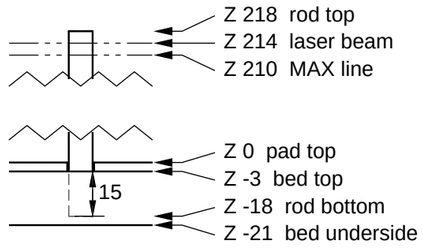
**PLAN (1:5): bed, pads, rods, station mount, gate posts; base rails and feet hidden**




**DETAIL C (1:2): one cell's rod holes = drill_template**
(cell centre at 0, 0; the machine drills them, BUILD_GUIDE 5.6)




**DETAIL D (1:2): rod in the bed,**
front, broken



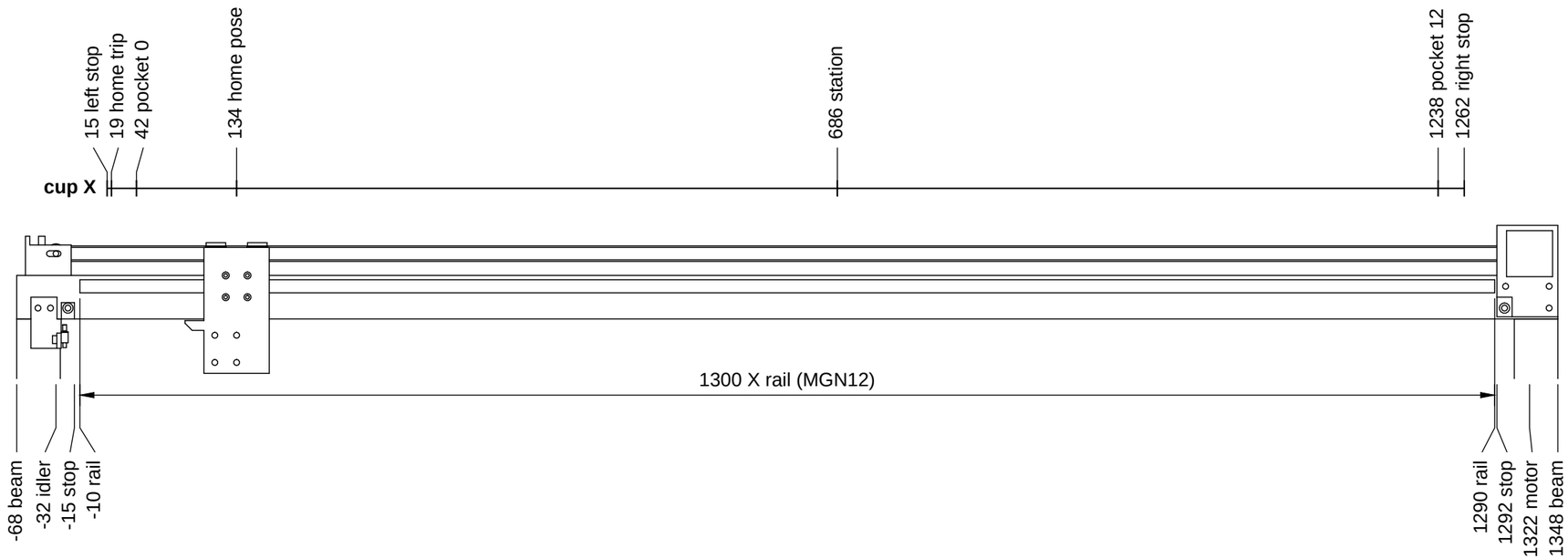

**Bed and pads**

| Item | Size (from the CAD) | Note |
|---|---|---|
| Bed, plywood (W) | 1416 x 289.5 x 18 | cut on site; test-drill for voids first (BUILD_GUIDE 5.1) |
| Pads (EVA + microfiber), 13 | 83 x 138 x 3 | spray adhesive, after the rods (5.9) |
| Rods (dowels), 82 | D-02 item C-11 | glued; plumb within 1 over 200 [Guessing target] (5.7) |

Datum O = machine origin, X 0 and Y 0 (the pocket-row centre line), as the software uses them; X ordinates from O. The rod holes are NOT laid out from the bed edges: the machine drills them through drill_template, so beam skew and belt stretch are built in (BUILD_GUIDE 5.6). Station mount: cell 7, four #8 wood screws, no pad. Gate posts: bare plywood, no shim (5.9).

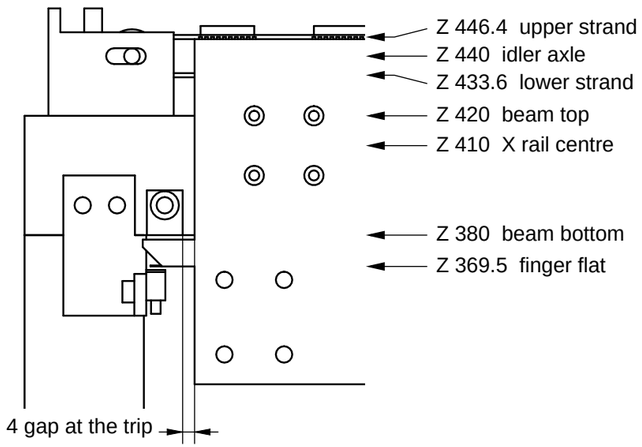
ROD HOLES: 82x Ø8 (CAD ROD_D; drill with the brad-point bit that matches the dowels: 5/16 in for the BOM's 7.94 dowels), 15 deep (ROD_EMBED), square to the bed (guide block, BUILD_GUIDE 5.6). Side rods on the cell wall lines, first X -4, then every 92, at Y ±57.5; front/back rods in pairs 60 apart about each pocket centre, at Y ±73.5; none front/back in the station cell 7.

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-03 of D-17** |
| **Bed, pockets and guide rods** | SCALE 1:2, 1:5 (as noted) |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |




**FRONT (1:5)**

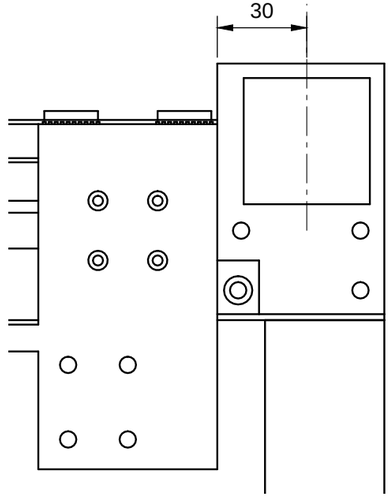
**DETAIL A (1:2): left end, carriage at the home-switch trip**
(cup X = X_HOME_SWITCH; front view)




Click at cup X 19 (+/-2, BUILD_GUIDE 6.8)

FRONT: X axis with the carriage at home (cup over pocket 1); Z axis and head not shown. cup X = machine X of the suction-cup axis. Belt: GT2 loop over the beam top, both ends clamped on the x_bracket tab (BUILD_GUIDE 6.5).

**DETAIL B (1:2): right end, carriage on the right stop**
(cup X = X_HARD_STOP[1]; front view)




x_bracket lands on the mount's left end

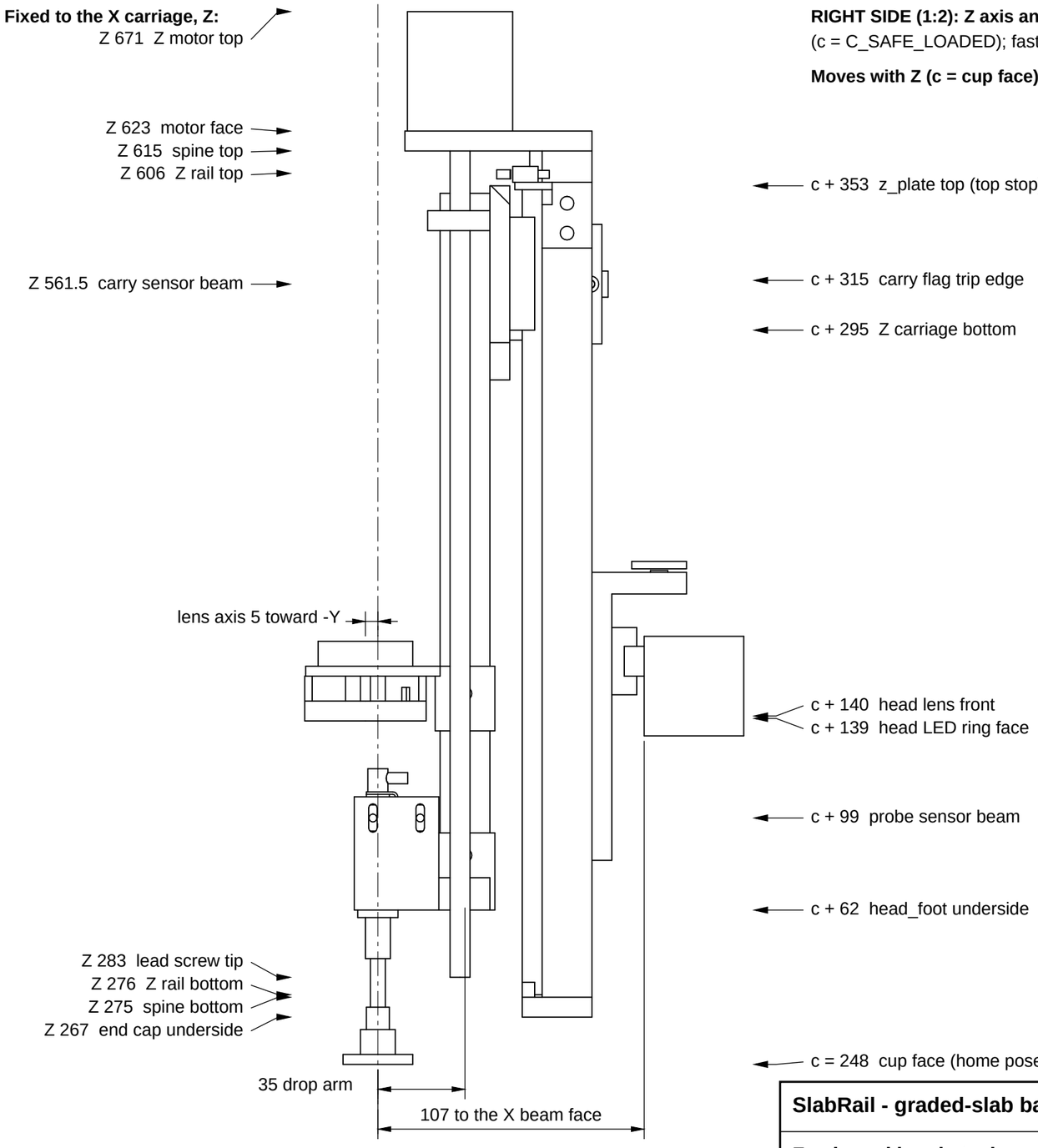
**Y stack-up, cup axis to the X beam's front face**

| Constant (layout.py) | mm | Part |
|---|---|---|
| ARM_TO_CUP | 35 | drop-arm centre to cup axis |
| EXT20 / 2 | 10 (of 20) | drop arm (2020): half = 10 |
| Z_PLATE_T | 8 | z_plate |
| MGN12_H | 13 | Z carriage (MGN12H) |
| EXT20 | 20 | Z spine (2040, 20 deep) |
| X_BRACKET_T | 8 | x_bracket |
| MGN12_H | 13 | X carriage (MGN12H) |
| = Y_CUP_TO_BEAM | 107 | sum: cup axis to beam front face |

CAD check: beam front face at Y 107

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-04 of D-17** |
| **X axis** | SCALE 1:2, 1:5 (as noted) |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

**Fixed to the X carriage, Z:**




**RIGHT SIDE (1:2): Z axis and head, home pose**
(c = C_SAFE_LOADED); fasteners not shown

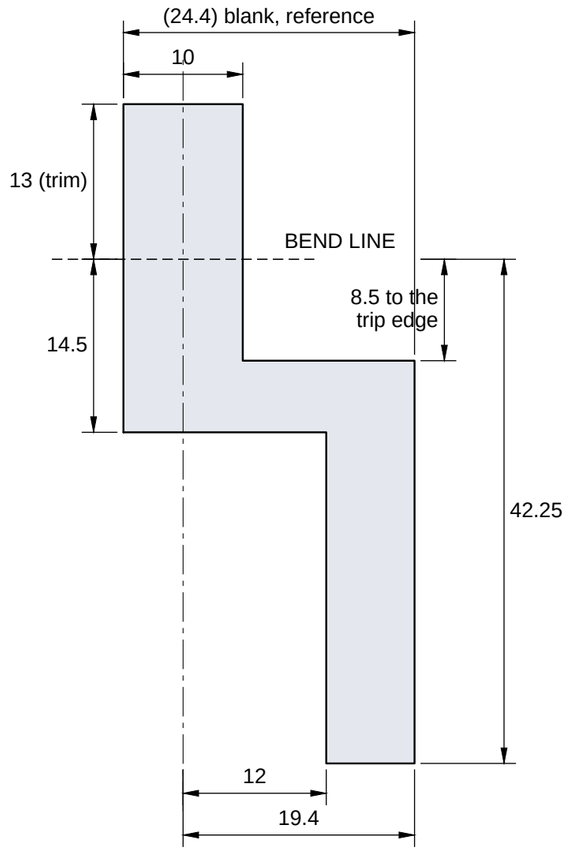
**Moves with Z (c = cup face):**

**Z-height table (values read from cad/layout.py)**

| Constant | Value | Meaning |
|---|---|---|
| SLAB_T | 6.6 | c at pick contact = top of the stack: one slab (pose pick_bottom) = SLAB_T; n slabs = n x SLAB_T, nominal |
| CUP_HANG | 2 | the free cup hangs this far below c; the probe trips after this much buffer travel |
| PROBE_TRIP_TOL | 0.5 | trip tolerance on CUP_HANG (BUILD_GUIDE 7.6) |
| PROBE_WINDOW | 12 | max probing travel past the predicted contact |
| BUFFER_STROKE | 20 | spring buffer stroke (SMC ZPT25UUK20-B5-A10) |
| STATION_LENS_Z | 23 | station lens front above the pad top |
| STATION_READ_C | 171.6 | c at the station read = STATION_LENS_Z + CAM_TO_SLAB + SLAB_T + CUP_HANG |
| CAM_TO_SLAB | 140 | both cameras: lens front to the slab face |
| HEAD_CAM_DY | -5 | head lens axis, Y from the cup axis (negative = toward the label end) |
| POST_H | 218 | guide-rod tops above the pad top |
| C_SAFE_LOADED | 248 | carry height, loaded and empty (C_SAFE_EMPTY is the same) |
| CARRY_SENSOR_C0 | 246.5 | carry sensor: beam blocked from this c up to the top stop |
| CARRY_BAND | 1 | carry sensor band: within it either reading is allowed |
| C_MAX | 250 | top of the software's Z travel |
| Z_SWITCH_C | 258 | Z top (home) switch trip |
| C_HARD_TOP | 262 | top hard stop: z_plate meets z_motor_mount |
| C_MIN | -12 | lowest c the design allows |
| C_HARD_BOTTOM | -14 | bottom hard stop: Z carriage skirts on the z_end_cap lips |
| Z_CARRIAGE_ABOVE_CUP | 295 | Z carriage bottom edge above the cup face |
| Z_SPINE_BOTTOM | 275 | spine bottom end above the pad top (set +/-1, BUILD_GUIDE 7.1) |
| Z_SCREW_DX | 24 | lead screw axis, X from the cup axis |
| Z_SPINE_OFFSET_X | -10 | Z spine centre, X from the cup axis |

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-05 of D-17** |
| **Z axis and head stack-up** | SCALE 1:2 |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

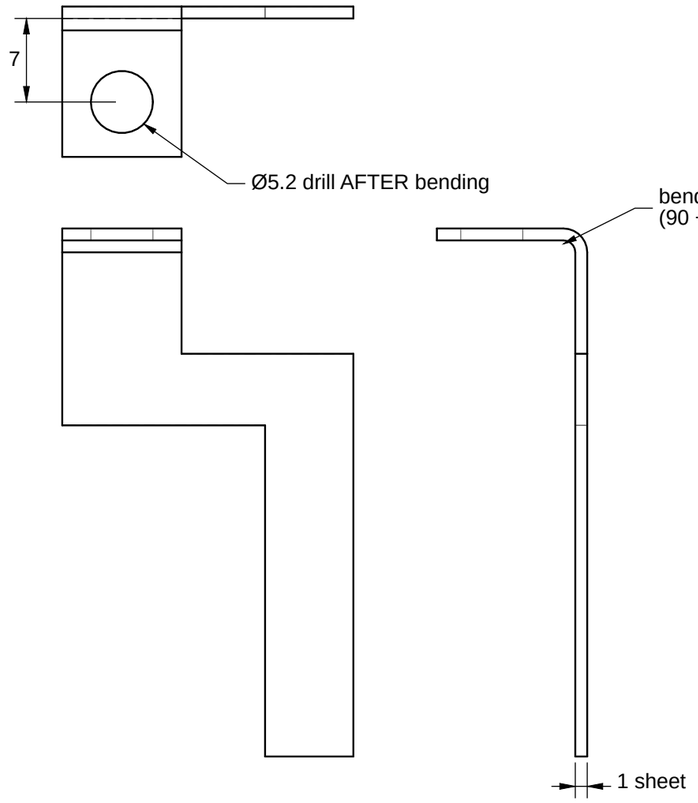
**PROBE FLAG BLANK (2:1), flat, before bending**
u from the neck centre line (= hole line), v from the bend line



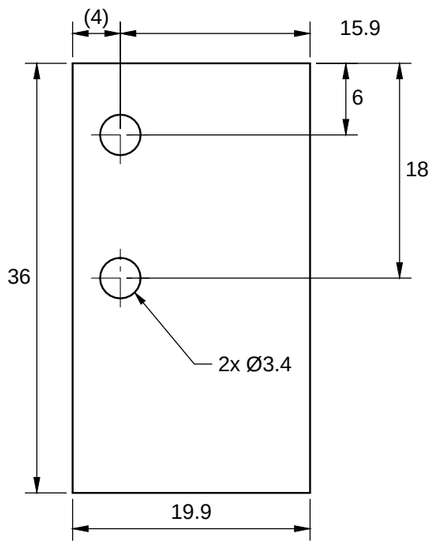

**Probe flag (BUILD_GUIDE 7.6; build.py probe_flag_blank.svg notes)**

Aluminium sheet, 1 thick (cut_list C-07). 1. Cut and file to the outline; the blade's top edge is the trip edge: straight, square. 2. This view shows the face toward the buffer. Neck in the vise, jaws at the BEND line; bend the tab 90° TOWARD YOU. 3. THEN drill on the centre line, 7 from the plate's inner face. 4. Trim the tab to 2 past the hole. 5. Deburr: it rides in head_foot's guide slot and the sensor slot. Outline: cad/out/probe_flag_blank.dxf (1:1 print: .svg).

**PROBE FLAG AS BENT (2:1), for reference**
(head frame: X along the row, Z above the cup face)




**CARRY FLAG (2:1), flat**
(seen from the front; top edge = trip edge)




**Carry flag (BUILD_GUIDE 7.4a)**

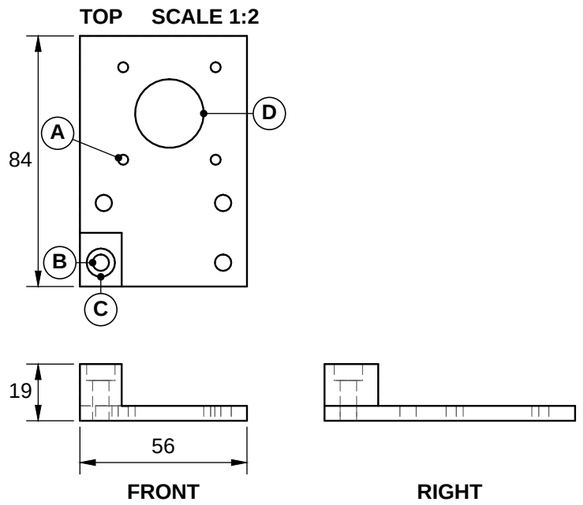
Aluminium sheet, 1 thick, flat (cut_list C-10). File the top edge straight. Fix to carry_flag_arm's leg end with 2 x M3x8, inner edge toward the spine, top edge flush with the arm's top.

Tolerance: outlines PROPOSED +/-0.2 (no doc states one); bend 90 +/-0.5° (BUILD_GUIDE 7.6). The trip points are set by adjustment on assembly (probe 7.6, carry 7.4a), not by these outlines.

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-06 of D-17** |
| **Flat parts (probe flag, carry flag)** | SCALE 2:1 |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

## x_motor_mount

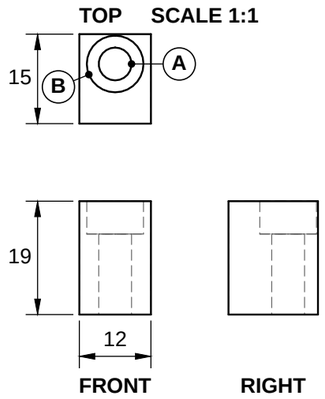
Printed part (P), qty 1, cad/out/printed/x_motor_mount.step. Print: PETG, beam face down (as exported), 5 walls, 40% infill




| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | 4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance |
| B | 4x Ø5.5 THRU, axis Z, centres spread over 41 x 20 (X x Y), not a grid: M5 clearance |
| C | Ø9.5 x 5.5 deep, axis Z: M5 head, sunk |
| D | Ø23 THRU, axis Z: NEMA 17 pilot clearance |
| E | Left end = right X hard stop face, 30 from the motor axis; motor axis 20 above the beam top |

## x_bumper_L

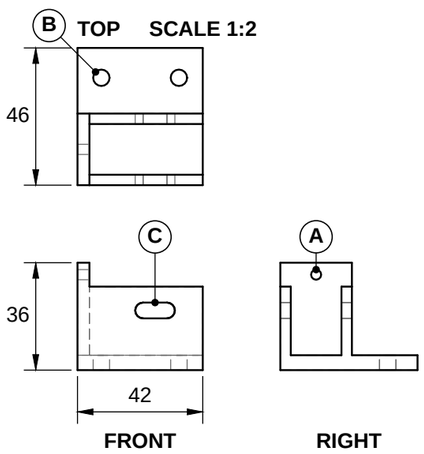
Printed part (P), qty 1, cad/out/printed/x_bumper_L.step. Print: PETG, beam face down (as exported), 4 walls, 40% infill




| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | Ø5.5 THRU, axis Z: M5 clearance |
| B | Ø9.5 x 5.5 deep, axis Z: M5 head, sunk |

## x_idler_mount

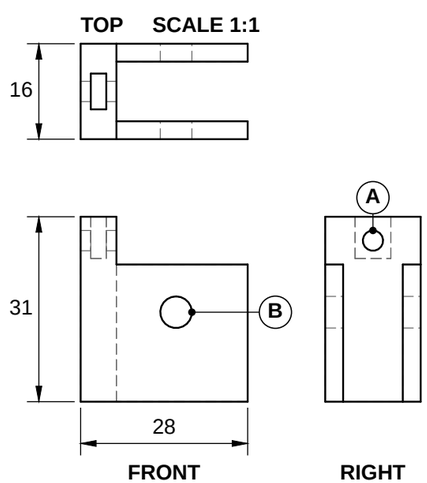
Printed part (P), qty 1, cad/out/printed/x_idler_mount.step. Print: PETG, foot down, 5 walls, 40% infill




| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | Ø3.4 THRU, axis X: M3 clearance |
| B | 2x Ø5.5 THRU, axis Z, 26 apart in X: M5 clearance |
| C | 2x slot 5.3 wide x 13.3 long, axis Y: M5 idler axle (tension travel) |

## x_idler_slider

Printed part (P), qty 1, cad/out/printed/x_idler_slider.step. Print: PETG, as exported, 5 walls, 50% infill



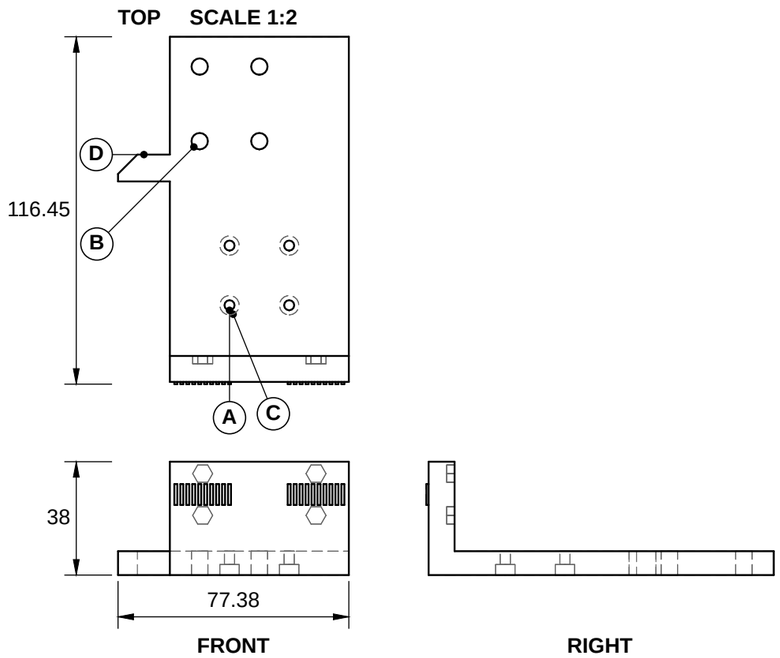

| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | Ø3.4 THRU (2 walls on one axis), axis X: M3 clearance |
| B | Ø5.3 THRU (2 walls on one axis), axis Y: M5 axle |
| C | M3 nut slot 2.6 wide, from the top, in the bridge |

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-08 of D-17** |
| **Printed parts: X drive** | SCALE 1:1, 1:2 (as noted) |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

## x_bracket

Printed part (P), qty 1, cad/out/printed/x_bracket.step. Print: PETG, print flat on its spine face (as exported), 6 walls, 50% infill; the finger off its left edge works the X home switch

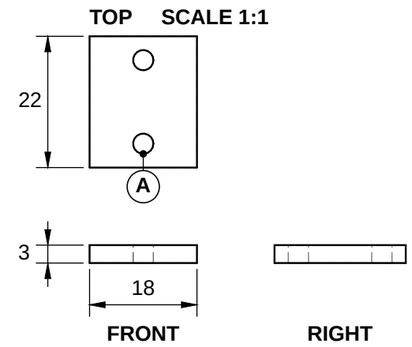



| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | 4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance |
| B | 4x Ø5.5 THRU, axis Z, on 20 x 25 (X x Y): M5 clearance |
| C | 4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore |
| D | X-switch finger 17.38 past the left edge; flat underside at machine Z 369.5; 45° lead-in 6.5 deep |
| E | GT2 rack on the tab floor, both sides: teeth at 2 pitch (belt teeth down, belt_clamp cap on top) |

## belt_clamp

Printed part (P), qty 2, cad/out/printed/belt_clamp.step. Print: PETG, flat, 100% infill

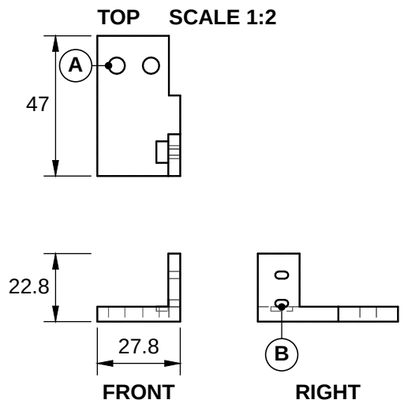
| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | 2x Ø3.4 THRU, axis Z, 14 apart in Y: M3 clearance |




## x_switch_mount

Printed part (P), qty 1, cad/out/printed/x_switch_mount.step. Print: PETG, front plate down (as exported), 4 walls, 40% infill; X home switch on 2 x M2x12, 2 x M5x10 button heads

| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | 2x Ø5.5 THRU, axis Z, 11.5 apart in X: M5 clearance |
| B | 2x slot 2.4 wide x 4.4 long, axis X, centres 9.5 apart in Z: M2 |

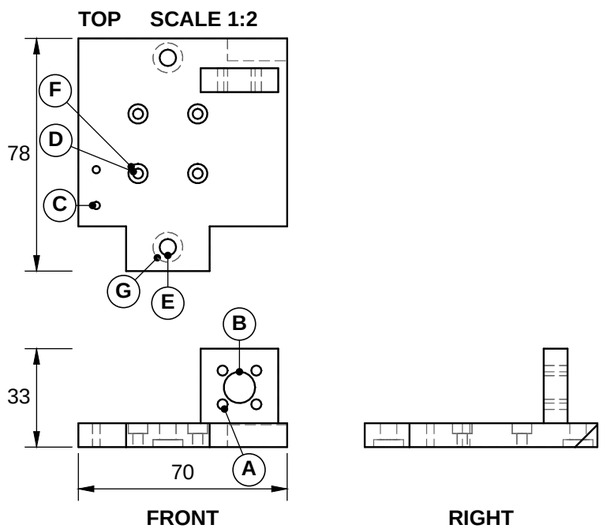



Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-09 of D-17** |
| **Printed parts: X carriage** | SCALE 1:1, 1:2 (as noted) |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

## z_plate

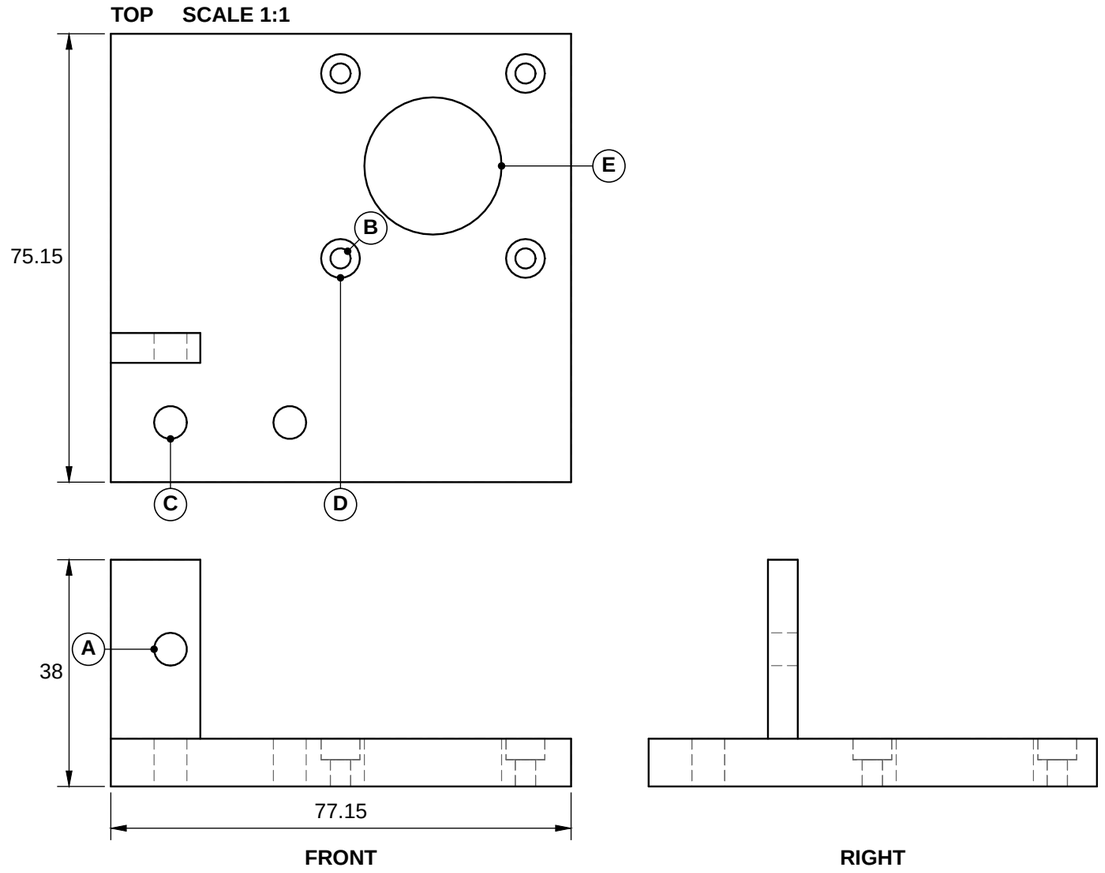
Printed part (P), qty 1, cad/out/printed/z_plate.step. Print: PETG, carriage face down, nut shelf up, 6 walls, 50% infill; the switch chamfer at the top right starts at the bed (45 deg)




| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | 4x Ø3.4 THRU, axis Y, on 11.31 x 11.31 (X x Z): M3 clearance |
| B | Ø10.5 THRU, axis Y: T8 nut body clearance |
| C | 2x Ø2.5 THRU, axis Z, 12 apart in Y: carry_flag_arm, M3x10 self-tapping |
| D | 4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance |
| E | 2x Ø5.5 THRU, axis Z, 63.5 apart in Y: M5 clearance |
| F | 4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore |
| G | 2x Ø10 x 2.5 deep, axis Z, 63.5 apart in Y: M5 button-head recess |
| H | Z-switch chamfer 45°, 7.5 x 7.5, 20 long at the right end of the top back edge |
| I | Top edge (Z top hard stop) 58 and T8 flange shelf top 48 above the Z carriage's bottom edge |

## z_motor_mount

Printed part (P), qty 1, cad/out/printed/z_motor_mount.step. Print: PETG, motor face down, tab up, 5 walls, 40% infill



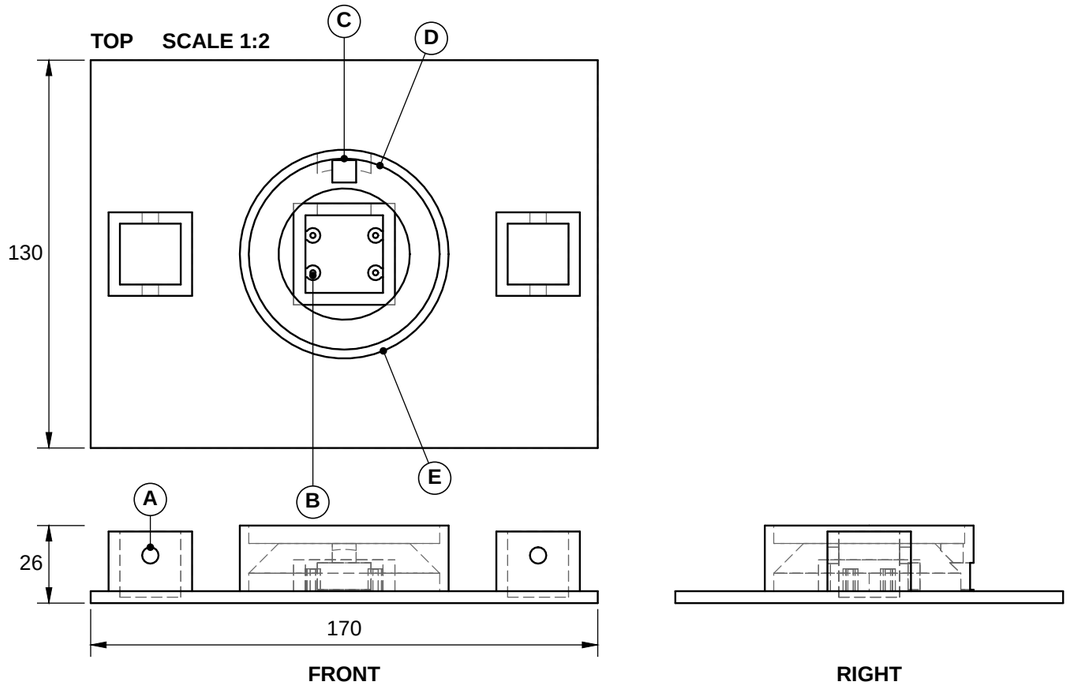

| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | Ø5.5 THRU, axis Y: M5 clearance |
| B | 4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance |
| C | 2x Ø5.5 THRU, axis Z, 20 apart in X: M5 clearance |
| D | 4x Ø6.5 x 3.5 deep, axis Z, on 31 x 31 (X x Y): M3 head counterbore |
| E | Ø23 THRU, axis Z: NEMA 17 pilot clearance |
| F | Plate underside sits on the spine top and is the Z top hard stop; tab hole 15 below it (left slot) |

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-10 of D-17** |
| **Printed parts: Z carriage plate, Z motor mount** | SCALE 1:1, 1:2 (as noted) |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

## phase0_base

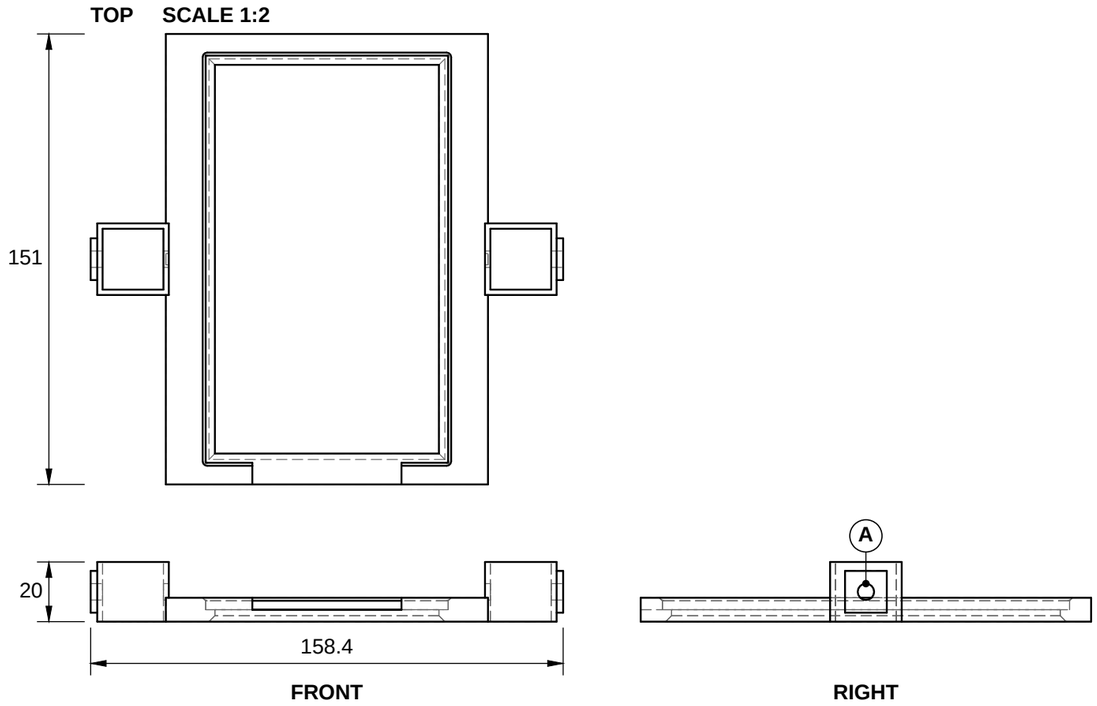
Printed part (P), qty 1, cad/out/printed/phase0_base.step. Print: PLA/PETG black (Phase 0 test jig)




| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | 2x Ø5.5 THRU (2 walls on one axis), axis Y, 130 apart in X: M5 clearance |
| B | 4x Ø1.8 x 7.5 deep, axis Z, on 21 x 12.5 (X x Y): camera board, M2 self-tapping |
| C | Ø64 x 6 deep, axis Z: hollow under the ring shelf (45° cone roof) |
| D | Ø64 x 6 deep, axis Z: station LED ring seat bore |
| E | Ø70 x 22 long (outside), axis Z: ring seat tube |
| F | Post sockets 20.4 square, floor 2 above the underside, centres 130 apart |
| G | Ring shelf 20 and socket cross holes 16 above the underside |

## phase0_tray

Printed part (P), qty 1, cad/out/printed/phase0_tray.step. Print: PLA/PETG (Phase 0 test jig), ledges down




| | FEATURES (axes = the STEP file's; positions: see the STEP) |
|---|---|
| A | Ø5.5 THRU (4 walls on one axis), axis X: M5 clearance |
| B | Slab recess 81.3 x 136.5, 4 deep; opening 75.1 x 130.3; the ledges carry the slab's front face |
| C | Post clamps 20.4 square, centres 130 apart |

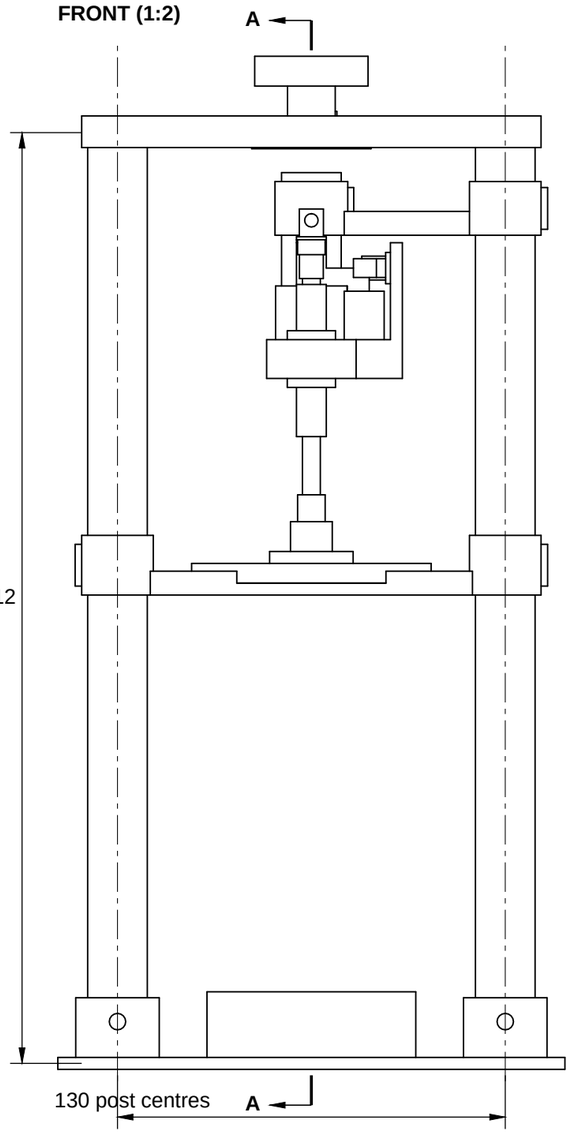
Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-15 of D-17** |
| **Printed parts: Phase 0 jig (base, tray)** | SCALE 1:2 |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |

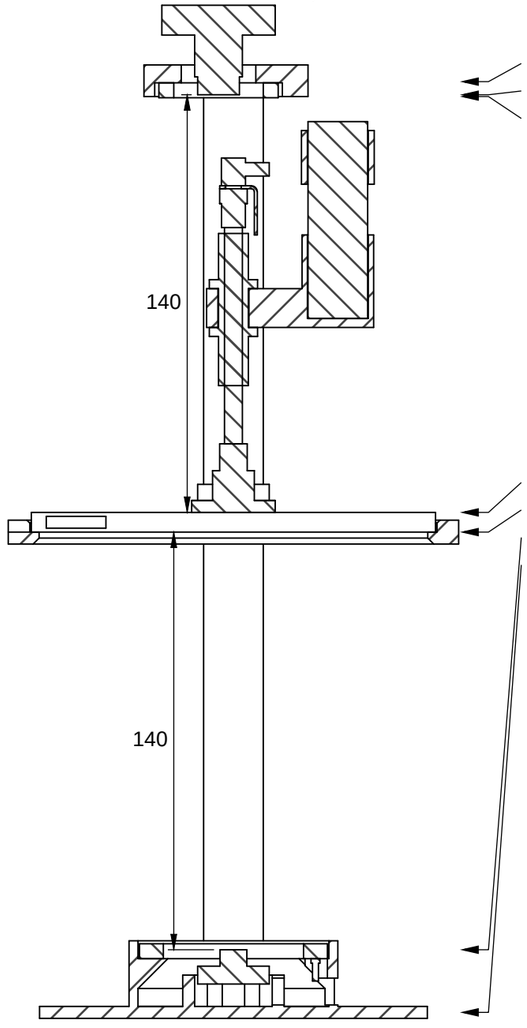
**PHASE 0A TEST JIG: one slab between the two cameras, as on the machine (BUILD_GUIDE 1.5); fasteners not shown**

**Heights above the base underside**

**FRONT (1:2)**



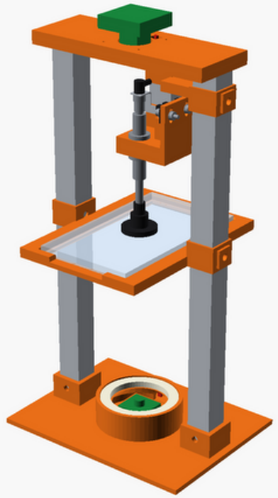

**SECTION A-A (1:2)**
on both lens axes, seen from +X; cut faces hatched




Z 314 post tops
Z 309.6 head lens front
Z 309 bridge underside

Z 169.6 slab back
Z 163 slab front face (on the ledges)
Z 23 station lens front
Z 2 post bottoms (socket floor)

**ISO (render; not to scale)**



| Part | Type | Qty | Sheet |
|---|---|---|---|
| phase0_base (camera mast, ring seat, post sockets) | P | 1 | D-15 |
| phase0_tray (the slab on edge ledges) | P | 1 | D-15 |
| phase0_top (bridge: head camera, head ring) | P | 1 | D-16 |
| phase0_occluder_mount | P | 1 | D-16 |
| head_foot, the machine's own (occluder) | P | 1 | D-12 |
| 2020 post, 312, cut square +/-0.5 (C-08) | C | 2 | D-02 |
| 2020 arm stub, 66 (C-09) | C | 1 | D-02 |
| probe flag (C-07) | S | 1 | D-06 |
| cameras, LED rings, SMC buffer + cup + elbow, endstop board | B | - | BOM.csv |

The posts set the bridge height: cut them square. Both lens-to-slab distances (RIGHT view) equal CAM_TO_SLAB = 140 on the machine (set +/-2, BUILD_GUIDE 1.5). Phase 0 is the go / no-go test before any motion parts are bought.

| | |
|---|---|
| **SlabRail - graded-slab batch scanner** | **SHEET D-17 of D-17** |
| **Phase 0 test jig** | SCALE 1:2 |
| Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication | UNITS: mm 3RD ANGLE |
| SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py) | DATE 2026-10-27 **DO NOT SCALE** |